

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater driver downstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Engine running > 10 second(s)
- Oxygen sensor heater active

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

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Diagnostic conditions

- Engine running > 10 second(s)
- Oxygen sensor heater active

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

P0050

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater driver for oxygen sensor upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Engine running > 10 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Oxygen sensor upstream of catalytic converter, bank 2 faulty

P0051

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater driver for oxygen sensor upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Engine running > 10 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Oxygen sensor upstream of catalytic converter, bank 2 faulty

P0052

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater driver for oxygen sensor upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Engine running > 10 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Oxygen sensor upstream of catalytic converter, bank 2 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater driver downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Engine running > 10 second(s)
- Oxygen sensor heater active

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater driver downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Engine running > 10 second(s)
- Oxygen sensor heater active

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater driver downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Engine running > 10 second(s)
- Oxygen sensor heater active

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

Diagnostic information - DME (DFI) control unit

Mass air flow in lower partial-load range implausibly high

Diagnostic conditions

- Driving with different operating conditions
- None of the following faults stored: Intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Possible fault causes

- ♦ Intake manifold pressure sensor - electric plug connection loose, oxidized, corroded, interrupted
- ♦ Tank vent valve faulty (stuck open)
- ♦ Fault in areas: intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment, variable intake manifold

Fault setting condition(s)

- Unacceptably large deviation between measured mass air flow and calculated mass air flow (see ⓘ).

ⓘ INFORMATION

This diagnostic function monitors the intake system between the throttle valve and engine (intake valves).

- ♦ A mass air flow is calculated from the signals from the throttle valve potentiometer, engine speed sensor, intake air temperature sensor and ambient pressure sensor as well as other engine operating values.
- ♦ The calculated mass air flow is compared with that measured by the pressure sensor.
- ♦ A deviation may be due to both the calculated mass air flow and the measured mass air flow.

Diagnostic information - DME (DFI) control unit

Ambient temperature sensor: deviation

(See Air-conditioning system regulator)

Diagnostic conditions

- ◆ Time since ignition off > 8 hours, then:
 - Drive > 25 km/h (16 mph) > 45 s within the first 6 minutes after engine start
 - NO drop in temperatures (e.g. due to drive from a garage into a cooler environment)

Possible fault causes

- ◆ Short circuit to B+/ground/between lines or open circuit in the lines
- ◆ Ambient temperature sensor faulty

Fault setting condition(s)

- The temperature measured by the ambient temperature sensor (**T050**) deviates too much from the mean value of the engine, engine oil, intake air and ambient air temperatures (**T160**)
(See **i**).

Fault effects

- ◆ Substitute value from model

i INFORMATION

Step 1

- ◆ When the engine is started after the ignition has been switched off for more than 8 hours, the engine coolant, engine oil, intake air and ambient air temperatures are compared with each other.
- ◆ They may deviate from the calculated mean value (T160) by only a certain amount (depending on the switch-off time).
- ◆ A suspected fault is set if the deviation is too large. However, the fault must still be confirmed in the second step.

Step 2

- ◆ Within the first six minutes after ignition on, a check is performed to determine whether the following conditions were met:
 - Drive > 25 km/h (16 mph) for longer than 45 seconds in total
 - None of the four temperatures decreased too much (e.g. due to drive from a garage into a cooler environment)
- ◆ If all conditions were met, the fault is entered in the fault memory.
- ◆ If one of the conditions was NOT met, the diagnostic result is rejected and NO fault is entered in the fault memory.
- ◆ If the vehicle is NOT driven within 6 minutes after starting and the aforementioned conditions are met, the suspected fault is rejected and NO fault is entered in the fault memory.

P0072

Diagnostic information - DME (DFI) control unit

Ambient temperature via CAN

(See Air-conditioning system regulator)

Diagnostic conditions

- Ignition on > 5 second(s)

Possible fault causes

- ♦ Outside temperature sensor fault
- ♦ The expected signal or message deviates from the specified value.

Fault effects

- ♦ Substitute value from model

P0073

Diagnostic information - DME (DFI) control unit

Ambient temperature via CAN

(See Air-conditioning system regulator)

Diagnostic conditions

- Ignition on > 5 second(s)

Possible fault causes

- ♦ Outside temperature sensor fault
- ♦ The expected signal or message deviates from the specified value.

Fault effects

- ♦ Substitute value from model

P0074

Diagnostic information - DME (DFI) control unit

Ambient temperature via CAN

(See Air-conditioning system regulator)

Diagnostic conditions

- Ignition on > 5 second(s)

Possible fault causes

- ♦ Outside temperature sensor fault
- ♦ The expected signal or message deviates from the specified value.

Fault effects

- ♦ Substitute value from model

Diagnostic information - DME (DFI) control unit

Intake valve lift control driver, bank 1

Diagnostic conditions

- Idle speed > 10 second(s)
- Then increase engine speed until the large valve lift has been switched (see Actual value [A220](#))

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Fuse faulty
- ♦ Solenoid valve for hydraulic valve lift control faulty

Diagnostic information - DME (DFI) control unit

Intake valve lift control driver, bank 1

Diagnostic conditions

- Idle speed > 10 second(s)
- Then increase engine speed until the large valve lift has been switched (see Actual value [A220](#))

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Fuse faulty
- ♦ Solenoid valve for hydraulic valve lift control faulty

Diagnostic information - DME (DFI) control unit

Exhaust valve lift control driver, bank 1

Diagnostic conditions

- Idle speed > 10 second(s)
- Then increase engine speed until the large valve lift has been switched (see Actual value [A225](#))

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Fuse faulty
- ♦ Solenoid valve for hydraulic valve lift control faulty

P0080

Diagnostic information - DME (DFI) control unit

Exhaust valve lift control driver, bank 1

Diagnostic conditions

- Idle speed > 10 second(s)
- Then increase engine speed until the large valve lift has been switched (see Actual value [A225](#))

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Fuse faulty
- ♦ Solenoid valve for hydraulic valve lift control faulty

Diagnostic information - DME (DFI) control unit

Intake valve lift control driver, bank 2

Diagnostic conditions

- Idle speed > 10 second(s)
- Then increase engine speed until the large valve lift has been switched (see Actual value [A220](#))

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Fuse faulty
- ♦ Solenoid valve for hydraulic valve lift control faulty

Diagnostic information - DME (DFI) control unit

Intake valve lift control driver, bank 2

Diagnostic conditions

- Idle speed > 10 second(s)
- Then increase engine speed until the large valve lift has been switched (see Actual value [A220](#))

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Fuse faulty
- ♦ Solenoid valve for hydraulic valve lift control faulty

Diagnostic information - DME (DFI) control unit

Exhaust valve lift control driver, bank 2

Diagnostic conditions

- Idle speed > 10 second(s)
- Then increase engine speed until the large valve lift has been switched (see Actual value [A225](#))

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Fuse faulty
- ♦ Solenoid valve for hydraulic valve lift control faulty

P0086

Diagnostic information - DME (DFI) control unit

Exhaust valve lift control driver, bank 2

Diagnostic conditions

- Idle speed > 10 second(s)
- Then increase engine speed until the large valve lift has been switched (see Actual value [A225](#))

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Fuse faulty
- ♦ Solenoid valve for hydraulic valve lift control faulty

Diagnostic information - DME (DFI) control unit

Fuel high pressure too low

INFORMATION

- ♦ Faults in the fuel high-pressure injection system area are frequently caused by defects in the pre-delivery circuit (fuel low pressure).
- ♦ Therefore, the pre-delivery circuit must be checked first in case of problems in the area of the fuel high-pressure injection system.

Diagnostic conditions

- Time since engine start > 5 second(s)
- Engine temperature - 10 ... + 100 °C (14 ... 212 °F)
- Engine running/driving

Possible fault causes (see Fault setting condition(s))

- ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
- ♦ Fuel tank (driven) empty
 - ▶ See Extended fault memory
 - ⇒ Actual value **40-Fuel Level Input** [0%]
- ♦ Quantity control valve faulty (in fuel high-pressure pump)
- ♦ Fuel high-pressure pump faulty
- ♦ Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see  below)

Fault setting condition(s)

- Average fuel high pressure continuously below a lower limit

INFORMATION

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
 - ♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%
 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

-  The inlet of the respective rail is constricted by a restrictor, which is used to dampen

pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure increase that is too slow, or a rapid drop in pressure when the fuel injectors open.
- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

Diagnostic information - DME (DFI) control unit

Fuel high pressure too high

Diagnostic conditions

- Time since engine start > 5 second(s)
- Engine temperature - 10 ... + 100 °C (14 ... 212 °F)
- Engine running/driving

Possible fault causes (see Fault setting condition(s))

- ♦ Quantity control valve faulty (in fuel high-pressure pump)
- ♦ Fuel high-pressure pump faulty
- ♦ Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see ⓘ below)

Fault setting condition(s)

- Average fuel high pressure continuously above an upper limit

ⓘ INFORMATION

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
 - ♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%
 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

ⓘ The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure increase that is too slow, or a rapid drop in pressure when the fuel injectors open.
- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

Diagnostic information - DME (DFI) control unit

Fuel low pressure implausible

Diagnostic conditions

- Engine running > 2 second(s)
- Engine temperature 60 ... 110 °C (140 ... 230 °F)

Possible fault causes

- ♦ Fuel tank (driven) empty
 - ▶ See Extended fault memory
 - ⇒ Actual value **40-Fuel Level Input** [0%]
- ♦ Fuel filter clogged up
- ♦ Fuel line kinked/trapped
- ♦ Pressure relief valve in fuel delivery unit leaking
- ♦ Fuel delivery unit leaking
- ♦ Electric fuel pump faulty
- ♦ Fuel low-pressure sensor out of adjustment

Fault setting condition(s)

Setpoint-to-actual comparison of fuel pump control unit activation.

- ♦ If the actual fuel low-pressure value cannot follow the setpoint or cannot follow it quickly enough, a fault entry is made for a deviation of the actual fuel low-pressure value of:
 - > 1,200 hPa below the setpoint for > 15 seconds
- or
- > 2,400 hPa above the setpoint for > 15 seconds

P008A

Diagnostic information - DME (DFI) control unit

Fuel low-pressure system pressure too low

Diagnostic conditions

- Engine running > 2 second(s)

Possible fault causes

- ♦ Fuel tank (driven) empty
 - ▶ See Extended fault memory
 - ⇒ Actual value **40-Fuel Level Input** [0%]
- ♦ Fuel line detached
- ♦ Electric fuel pump faulty
- ♦ Fuel low-pressure sensor faulty

Fault setting condition(s)

Actual fuel low pressure value below 3,000 hPa minimum pressure for 0.7 seconds.

P008B

Diagnostic information - DME (DFI) control unit

Fuel low-pressure system pressure too high

Diagnostic conditions

- Engine running > 2 second(s)

Possible fault causes

- ♦ Fuel low-pressure sensor faulty
- ♦ Fuel high-pressure pump faulty (supplies high pressure to low-pressure side)

Fault setting condition(s)

Actual fuel low pressure value above 9,000 hPa maximum pressure for 0.7 seconds.

- ⓘ The pressure relief valve in the fuel delivery unit opens at approx. 7,000 hPa.

P0090

Diagnostic information - DME (DFI) control unit

Quantity control valve output stage

Diagnostic conditions

- Ignition on > 5 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Quantity control valve faulty (in fuel high-pressure pump)

P0091

Diagnostic information - DME (DFI) control unit

Quantity control valve output stage

Diagnostic conditions

- Ignition on > 5 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Quantity control valve faulty (in fuel high-pressure pump)

P0092

Diagnostic information - DME (DFI) control unit

Quantity control valve output stage

Diagnostic conditions

- Ignition on > 5 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Quantity control valve faulty (in fuel high-pressure pump)

Diagnostic information - DME (DFI) control unit

Engine coolant shutoff valve stuck closed
or
Thermostat stuck closed

Diagnostic conditions

- Uniform driving (only slight changes in rpm and torque)
- Setpoint engine temperature (T005) > 104.25 °C (219.65 °F)
- **Engine coolant temperature** (T020) = setpoint engine temperature (T005)
- **Coolant temperature** at radiator outlet (T025) < 80 °C (176 °F)
- **Engine speed** 1,200 ... 3,500 rpm
- Air conditioner OFF (= heating valve must not switch during diagnostics)

Possible fault causes

- ♦ Solenoid switchover valve for coolant shutoff valve vacuum unit mechanically stuck
- ♦ Vacuum lines on the solenoid switchover valve interchanged
- ♦ Vacuum lines or electrical lines of the solenoid switchover valve interchanged with another solenoid switchover valve
- ♦ Engine coolant shutoff valve faulty (stuck closed)
- ♦ Thermostat faulty (stuck closed)
- ♦ Air trapped in the cooling system (e.g. after repairs were performed on the cooling system)
- ♦ Incorrect measurement by engine temperature sensor or radiator return temperature sensor, but still in the plausible range (-40...+150 °C/-40...+302 °F)

Diagnostic function ⇒ **See function description**

P0105

Diagnostic information - DME (DFI) control unit

Intake manifold pressure sensor faulty

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Intake manifold pressure sensor faulty

Fault effects

- ♦ Substitute value from model

P0106

Diagnostic information - DME (DFI) control unit

Intake manifold pressure sensor implausible

Diagnostic conditions

- Ignition on > 10 second(s)
- then: Acceleration with wide-open throttle for as long as possible, immediately followed by an overrun phase lasting as long as possible
- None of the following faults stored: P1638, P1639, P0107, P0108, P2228, P2229, P1183, P1184, P1187, P1188, P1189, P1190

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Ground supply faulty
- ♦ Intake manifold pressure sensor faulty
- **ONLY** for '**40-Time after Start**' greater than **0** (see Extended fault memory):
 - ♦ Throttle valve adjusting unit soiled

Fault effects

- ♦ Substitute value from model

P0107

Diagnostic information - DME (DFI) control unit

Intake manifold pressure sensor: electrical fault

Diagnostic conditions

- Ignition on or engine running

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Intake manifold pressure sensor faulty

Fault effects

- ♦ Substitute value from model

P0108

Diagnostic information - DME (DFI) control unit

Intake manifold pressure sensor: electrical fault

Diagnostic conditions

- Ignition on or engine running

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Intake manifold pressure sensor faulty

Fault effects

- ♦ Substitute value from model

Diagnostic information - DME (DFI) control unit

Intake air temperature implausible during cold start

Diagnostic conditions

- Time since ignition off > 8 hours, then:
 - Drive > 25 km/h (15 mph) > 45 s within the first 6 minutes after engine start
 - NO drop in temperatures (e.g. due to drive from a garage into a cooler environment)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Intake air temperature sensor faulty

Fault setting condition(s)

- The temperature measured by the intake air temperature sensor (T040) deviates too much from the mean value of the engine, engine oil, intake air and ambient air temperatures (T160) (see ⓘ)

Fault effects

- ♦ Substitute value from model

ⓘ INFORMATION

Step 1

- ♦ When the engine is started after the ignition has been switched off for more than 8 hours, the engine coolant, engine oil, intake air and ambient air temperatures are compared with each other.
- ♦ They may deviate from the calculated mean value (T160) by only a certain amount (depending on the switch-off time).
- ♦ A suspected fault is set if the deviation is too large. However, the fault must still be confirmed in the second step.

Step 2

- ♦ Within the first six minutes after ignition on, a check is performed to determine whether the following conditions were met:
 - Drive > 25 km/h (15 mph) for longer than 45 seconds in total
 - None of the four temperatures decreased too much (e.g. due to drive from a garage into a cooler environment)
- ♦ If all conditions were met, the fault is entered in the fault memory.
- ♦ If one of the conditions was NOT met, the diagnostic result is rejected and NO fault is entered in the fault memory.
- ♦ If the vehicle is NOT driven in compliance with the aforementioned conditions within 6 minutes after starting, the suspected fault is rejected and NO fault is entered in the fault memory.

P0111

Diagnostic information - DME (DFI) control unit

Intake air temperature implausible

Diagnostic conditions

- Vehicle has NOT been and is NOT standing in the sun.
- Engine off > 9 hour(s)
- Engine coolant temperature cooling > 35 °C (95 °F) with engine off
- Then cold start at:
 - Engine coolant temperature < 37 °C (98.6 °F)
 - Difference between intake air temperature <=> engine coolant temperature < 6 °C (42.8 °F)
- Then driving, with:
 - Engine running > 23 minute(s)
 - Distance driven > 7 km (4 mls)
 - Engine coolant temperature up to > 94 °C (201 °F)
- Then idling > 2 minute(s)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Intake air temperature sensor faulty (in intake manifold pressure sensor)
- ♦ Resistor installed (instead of intake air temperature sensor)

Fault setting condition(s)

- Change in intake air temperature less than 1.5 °C (34.7 °F) when diagnostic conditions are met
- or
- Intake air temperature values not within the permitted range

Fault effects

- ♦ Substitute value from model

P0114

Diagnostic information - DME (DFI) control unit

Intake air temperature sensor: electrical fault

Diagnostic conditions

- Ignition on > 10 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Intake air temperature sensor faulty

P0116

Diagnostic information - DME (DFI) control unit

Engine coolant temperature implausible

INFORMATION

The fault **P2081-Thermostat stuck open** is always prioritized during fault finding if it was entered in addition to fault P0116.

♦ The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

♦ **If only** fault **P0116** was entered, **check** which **DTC-DFCC** is entered in the extended fault memory.

► If **DTC-DFCC 00EE78** is entered, perform the **Thermostat system test before starting fault finding**

► see PIWIS Tester ⇒ DME ⇒ Maintenance/repairs ⇒ System tests

Diagnostic conditions

- Perform Thermostat system test

or

- Perform **First drive** with:
 - Engine start at (measured) engine temperature (**T020**) less than 30 °C (86 °F)
 - Drive for more than 20 minutes
- **Then drive the vehicle a second time** with:
 - Engine start again at (measured) engine temperature (**T020**) higher than 74 °C (165.2 °F)
 - Drive with engine speed greater than 1,300 rpm
 - High and low cooling power
 - Then with vehicle at a standstill, leave the engine running at idle speed for more than 45 seconds

Possible fault causes

 The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

- For **DTC-DFCC 00EE77**:
 - ♦ Engine coolant temperature sensor faulty
 - ♦ Resistor installed (instead of coolant temperature sensor)
- For **DTC-DFCC 00EE78**:
 - ♦ Thermostat faulty
 - ♦ Seal in thermostat housing faulty
 - ♦ Thermostat housing faulty
 - ♦ Air in the cooling system
 - ♦ Combination valve for cooling system bleeding not closed (service bow is up)
 - ♦ Combination valve for cooling system bleeding leaking
 - ♦ Engine coolant temperature sensor faulty